Baran Atalar

Ph.D. Candidate, Electrical and Computer Engineering, Carnegie Mellon University

batalar@andrew.cmu.edu · Google Scholar · Website · LinkedIn

Research Interests

Online learning and multi-armed bandits, optimization, and cost-efficient AI systems, with a focus on semantic caching and resource-aware serving of large language models (LLMs). More broadly, reinforcement learning, efficiency in LLM inference and multi-agent systems.

Education

Ph.D., Electrical and Computer Engineering — Aug 2022 – May 2027 (expected)
Carnegie Mellon University, Pittsburgh, USA
Advisor: Prof. Carlee Joe-Wong.

M.S., Electrical and Computer Engineering — Aug 2022 – Dec 2025
Carnegie Mellon University, Pittsburgh, USA
GPA 3.81/4.00. Earned during the Ph.D.

B.Sc., Electrical and Electronics Engineering — Sep 2018 – Jun 2022
Bilkent University, Ankara, Turkey
GPA 3.88/4.00. High Honor Student in every semester.

Honors & Awards

ACM SIGMOBILE Student Travel Grant, ACM MobiHoc 2026 — 2026
Awarded to support student attendance and presentation at ACM MobiHoc 2026.

NSF Student Travel Grant, IEEE INFOCOM 2026 — 2026
Awarded to support student attendance and presentation at IEEE INFOCOM 2026.

Carnegie Institute of Technology Dean's Fellowship, Carnegie Mellon University — 2022 – 2023
Funded the first year of Ph.D. studies.

Merit Scholarships, Bilkent University — 2019 – 2022
Awarded by rank in cohort (among 148 students): 100% (2021–22, top 1%), 80% (2020–21, top 2%), 60% (2019–20, top 3%).

Research Experience

Graduate Researcher — Aug 2022 – present
Carnegie Mellon University, advised by Prof. Carlee Joe-Wong

- Semantic caching over continuous query spaces (ACM MobiHoc 2026)
- Offline and online learning for semantic caching (IEEE INFOCOM 2026)
- Combinatorial neural bandits for recommendation systems (AAAI 2025)
- LLM selection for multi-step pipelines (preprint, under review)
- Multi-turn semantic caching in conversations (preprint, under review)

Undergraduate Researcher — Jan 2021 – Aug 2022
Bilkent University, advised by Assoc. Prof. Cem Tekin

- Contextual combinatorial GP bandits with group constraints (TMLR 2023)

Publications

* Equal contribution.

Conference Proceedings

[C3] **B. Atalar**, X. Liu, J. Zuo, S. Wang, W. Chen, and C. Joe-Wong. "Continuous Semantic Caching for Low-Cost LLM Serving." In *Proceedings of ACM MobiHoc* (to appear), Tokyo, Japan, 2026. doi:10.1145/3842721.3849656; arXiv:2604.20021.

[C2] X. Liu*, **B. Atalar***, X. Dai, J. Zuo, S. Wang, J. C. S. Lui, W. Chen, and C. Joe-Wong. "Semantic Caching for Low-Cost LLM Serving: From Offline Learning to Online Adaptation." In *Proceedings of IEEE INFOCOM*, pp. 1–10, 2026. doi:10.1109/INFOCOM59046.2026.11571467.

[C1] **B. Atalar** and C. Joe-Wong. "Neural Combinatorial Clustered Bandits for Recommendation Systems." In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 15, pp. 15417–15426, Philadelphia, PA, 2025. doi:10.1609/aaai.v39i15.33692.

Journal Articles

[J1] S. Elahi*, **B. Atalar***, S. Öğüt*, and C. Tekin. "Contextual Combinatorial Multi-output GP Bandits with Group Constraints." *Transactions on Machine Learning Research*, 2023.

Preprints

[P2] **B. Atalar**, S. Beyene, X. Liu, and C. Joe-Wong. "Multi-Turn Semantic Caching for Low-Cost LLM Serving." arXiv:2611.01077, 2026. Under review.

[P1] **B. Atalar**, E. Zhang, and C. Joe-Wong. "Neural Bandit Based Optimal LLM Selection for a Pipeline of Subtasks." arXiv:2508.09958, 2025. Under review.

Conference Presentations

IEEE INFOCOM; ACM MobiHoc, Tokyo, Japan (upcoming, November 2026) — 2026

ACM SIGMETRICS, Student Research Competition (accepted extended abstract) — 2025

AAAI Conference on Artificial Intelligence, Philadelphia, PA — 2025

Teaching

Teaching Assistant, *18-202 Mathematical Foundations of Electrical Engineering*, Carnegie Mellon University — Fall 2026

Guest Lecturer, "Exploration in RL," *TCSS 590 Theory and Algorithms of Reinforcement Learning*, University of Washington — Spring 2026

Teaching Assistant, *18-661 Introduction to ML for Engineers*, Carnegie Mellon University — Spring & Fall 2024

Professional Service

Reviewer, conferences: NeurIPS, AISTATS, AAAI, COLM, ACM SIGKDD, ECML PKDD.

Reviewer, journals: IEEE/ACM Transactions on Networking, ACM ToMPECS, IEEE TETCI.

Program Committee Member: AAAI

Skills

Programming: Python, MATLAB, SQL, LaTeX.

Tools: PyTorch, TensorFlow, PySpark, AWS, Jupyter.

Relevant Coursework

Carnegie Mellon University: Advanced Introduction to Machine Learning, Intermediate Statistics, Convex Optimization, Foundations of Reinforcement Learning, Machine Learning with Large Datasets, Generative AI, Algorithms in the Real World, Inference Algorithms for Language Modeling.

Selected Projects

API-based LLM Inference Server — 2025

Course project, 11-664 Inference Algorithms for Language Modeling, Carnegie Mellon University

Designed and implemented an inference server that optimizes latency and accuracy for graph shortest-path, medical Q&A and open-ended generation tasks.

Landing a Fixed-Wing UAV with Reinforcement Learning — 2021 – 2022

Capstone project with Titra Teknoloji A.Ş., Bilkent University

Implemented PPO, DDPG and TRPO methods for a fixed-wing UAV in Gazebo and GymFC.

Industry Experience

Summer Intern, ASELSAN A.Ş. — Summer 2021

Designed sparse antenna arrays with genetic algorithms and implemented antenna array factor computations in Python.

Summer Intern, Meteksan Savunma A.Ş. — Summer 2020

Implemented a PRBS receiver in VHDL.